

Numerical methods

Problems for the practices

Week 3 – Gaussian elimination and its applications

3.1. Solve the below simultaneous linear systems using Gaussian elimination (GE).

$$\begin{array}{cccccc|c} x_1 & - & x_2 & + & 2x_3 & + & x_4 & = & 3 \\ 2x_1 & + & 3x_2 & + & x_3 & - & x_4 & = & 5 \\ 3x_1 & + & 2x_2 & + & 3x_3 & & & = & 8 \\ x_1 & + & 4x_2 & - & x_3 & - & 2x_4 & = & 2 \end{array}$$

3.2. Use Gaussian elimination to find the inverse and the determinant of the matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

3.3. Find the inverse of the following matrix using GE.

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

3.4. We are given the linear system

$$\begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

with $0 < \varepsilon \ll 1$.

- (a) Solve the linear system using GE without row change.
- (b) Let $\varepsilon = 10^{-17}$, calculating with 16 significant digits (mantissa length $t = 16$) solve the linear system using GE without row change.
- (c) Solve the linear system using GE with (scaled) partial pivoting.

3.5. Let us consider

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 4 & 7 \\ 2 & 5 & 9 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

and solve the linear system $Ax = b$

- (a) using Gaussian elimination (without row/column interchange);

- (b) using Gaussian elimination with partial pivoting.
- (c) Find the determinant of A .
- (d) Find the inverse matrix of A using GE.

3.6. Solve the following $n \times n$ linear system using Gaussian elimination.

$$\left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 1 & 0 & 0 & & \vdots & 0 \\ -1 & 0 & 1 & 0 & & \vdots & \vdots \\ 0 & -1 & 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & -1 & 0 & 1 & 0 \end{array} \right]$$

3.7. Consider

$$A = \begin{bmatrix} 4 & 2 & 6 & -2 \\ 2 & 10 & 9 & 5 \\ 6 & 9 & 14 & 2 \\ -2 & 5 & 2 & 7 \end{bmatrix}.$$

Factor the matrix A into LU decomposition

- (a) using the lower triangular matrices L_i ($i = 1, 2, 3$);
- (b) using Gaussian elimination (compressed form);
- (c) using matrix multiplication (direct method).
- (d) Find the determinant of the matrix A .